

AY 2025/2026



POLITECNICO
MILANO 1863

Internet Of Things: Homework Exercise 1

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Version 1.0
May 28, 2026

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1 Assumptions

Assume that the number of vibration peaks detected in each 10-second window follows a Poisson distribution with average rate: $\lambda = 0.4$ peaks/window.

The system is operated with IEEE 802.15.4 in beacon-enabled mode, CFP only.

Assume:

$$R = 250 \text{ kbps}$$

$$L = 128 \text{ bytes}$$

where R is the nominal bit rate and L is the packet size.

Assume that:

- one packet fits exactly in one CFP slot
- the CFP is dimensioned according to the expected traffic load
- the reporting period is 10 seconds
- the monitoring system is composed of 1 PAN coordinator and 3 vibration monitoring nodes

2 Question 1

Compute the Probability Mass Function of the output rate: $P(r = r_0)$, $P(r = r_1)$, $P(r = r_2)$ where r_0 , r_1 , and r_2 are the output rates when the node detects 0, 1, or more than 1 vibration peaks, respectively.

We can use this formula to calculate the PMF

$$P(k) = \frac{e^{-\lambda} \cdot \lambda^k}{k!}$$

So

$$P(r = r_0) = \frac{e^{-0.4} \cdot 0.4^0}{0!} = 0.6703$$

$$P(r = r_1) = \frac{e^{-0.4} \cdot 0.4^1}{1!} = 0.2681$$

Given that the sum of all probabilities must be 1,

$$P(r = r_3) = 1 - P(r = r_1) - P(r = r_2) = 0.0616$$

3 Question 2

Based on the output rate PMF, compute a consistent CFP slot assignment for the system.

Here we must consider the worst-case scenario, the max payload is 4 KB, when there is more than 1 peak.

$$\begin{aligned}
 Slot_{PerNode} &= \lceil \frac{4 \text{ KB}}{128 \text{ B}} \rceil = \lceil \frac{4000 \text{ B}}{128 \text{ B}} \rceil = \lceil 31, 25 \rceil = 32 \\
 BI &= \frac{L}{r_{min}} = \frac{128 \text{ B}}{\frac{512 \text{ B}}{10 \text{ seconds}}} = 2.5 \text{ seconds} \\
 N_{Slots} &= \frac{Slot_{PerNode}}{N_{super frame}} \cdot N_{Nodes} = \frac{32}{\frac{10}{2.5}} \cdot 3 = 24
 \end{aligned}$$

4 Question 3

Compute:

- T_S
- N_{Slots}
- T_{Active}
- $T_{Inactive}$
- The duty cycle of the system η

$$\begin{aligned}
 T_S &= \frac{L}{R} = \frac{128 \text{ B}}{250 \text{ kbps}} = \frac{1024 \text{ bits}}{250000 \frac{\text{bits}}{\text{sec}}} = 4.096 \text{ ms} \\
 N_{Slots} &= 24 \\
 T_{Active} &= (1 + N_{Slots}) \cdot T_S = 25 \cdot 4.096 \text{ ms} = 102.4 \text{ ms} \\
 T_{Inactive} &= BI - T_{Active} = 2.5 \text{ s} - 102.4 \text{ ms} = 2.397 \text{ s} \\
 \eta &= \frac{T_{Active}}{BI} \cdot 100 = \frac{102.4 \text{ ms}}{2.5 \text{ s}} \cdot 100 = 4.096\%
 \end{aligned}$$

5 Question 4

How many additional vibration monitoring nodes can be added while keeping the duty cycle below 10%?

To keep the $\eta < 10\%$ we have a max

$$\begin{aligned}
 T_{Active} &= \frac{10\% \cdot BI}{100\%} = \frac{10\% \cdot 2.5 \text{ s}}{100\%} = 250 \text{ ms} \\
 Slots_{Available} &= \frac{T_{Active}}{T_S} = \frac{250 \text{ ms}}{4.096 \text{ ms}} = 61.035 \text{ slots} \\
 Max_{Nodes} &= \lfloor \frac{Slots_{Available}}{Slot_{PerSuperframe}} \rfloor = \lfloor \frac{61.035}{8} \rfloor = \lfloor 7.629 \rfloor = 7 \\
 \Delta Nodes &= Max_{Nodes} - 3 = 4 \text{ nodes}
 \end{aligned}$$